Public engagement and research software engineering – a vital role?

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epcc

What is a Research Software Engineer?

A **Research Software Engineer (RSE)** is a professional who combines expertise in **software development** with an understanding of **research processes and goals**.

Their role bridges the gap between traditional researchers (e.g., scientists, academics) and software engineers.

What does an RSE do?

- **Develops software for research** – writing code for simulations, data analysis, modeling, machine learning, etc.
- **Ensures software quality** – using good engineering practices: testing, version control, documentation.
- **Collaborates with researchers** – understanding scientific questions and designing software solutions to address them.
- **Supports reproducible science** – creating tools that make research easier to share, reproduce, and extend.
- **Optimizes performance** – making research code faster, more scalable, or able to run on high-performance computing (HPC) systems.

Why science needs more research software engineers

Ten years after their profession got its name, research software engineers seek to swell their ranks.

By Chris Woolston

What do RSEs do?

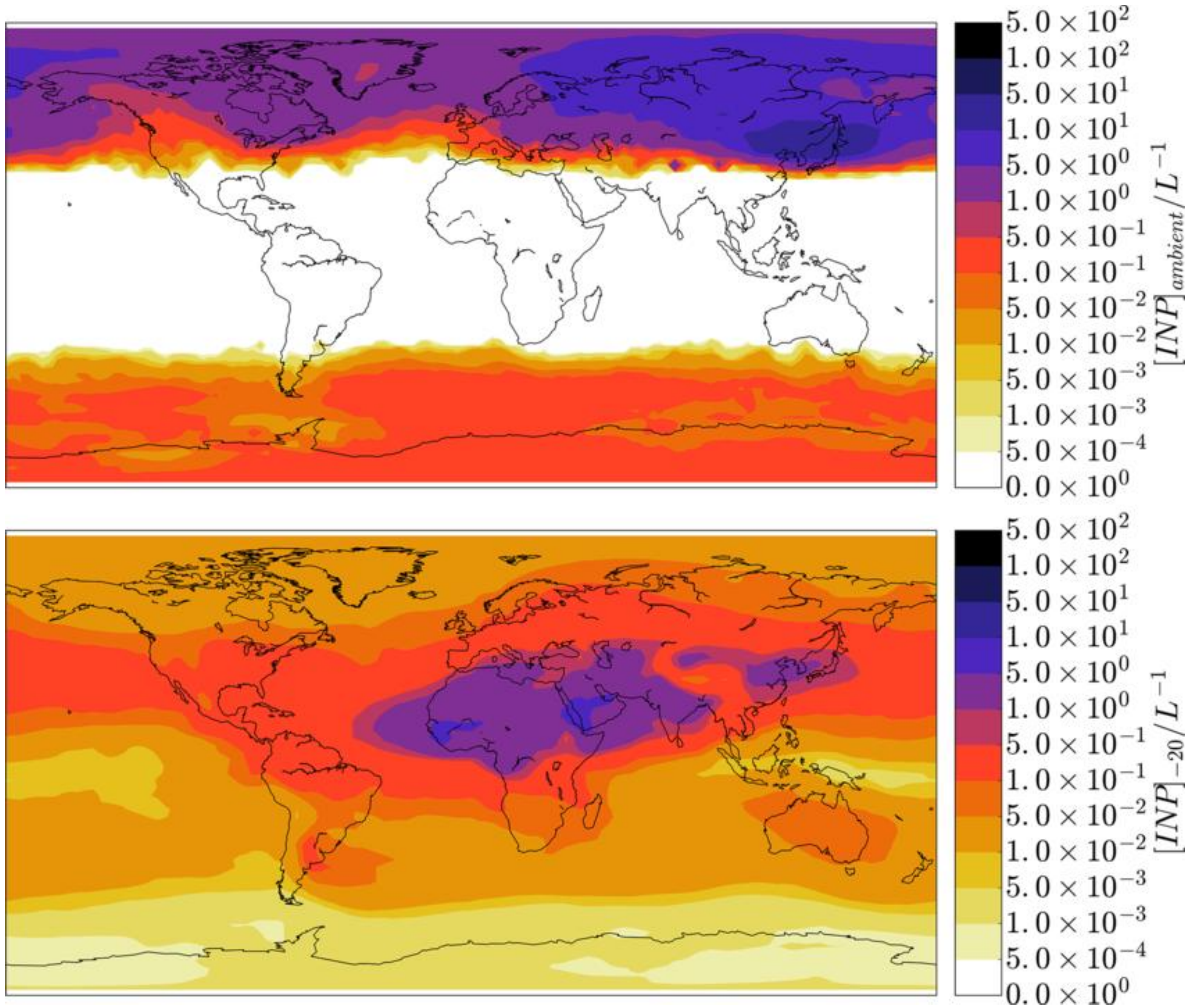
Fundamentally, RSEs build software to support scientific research.

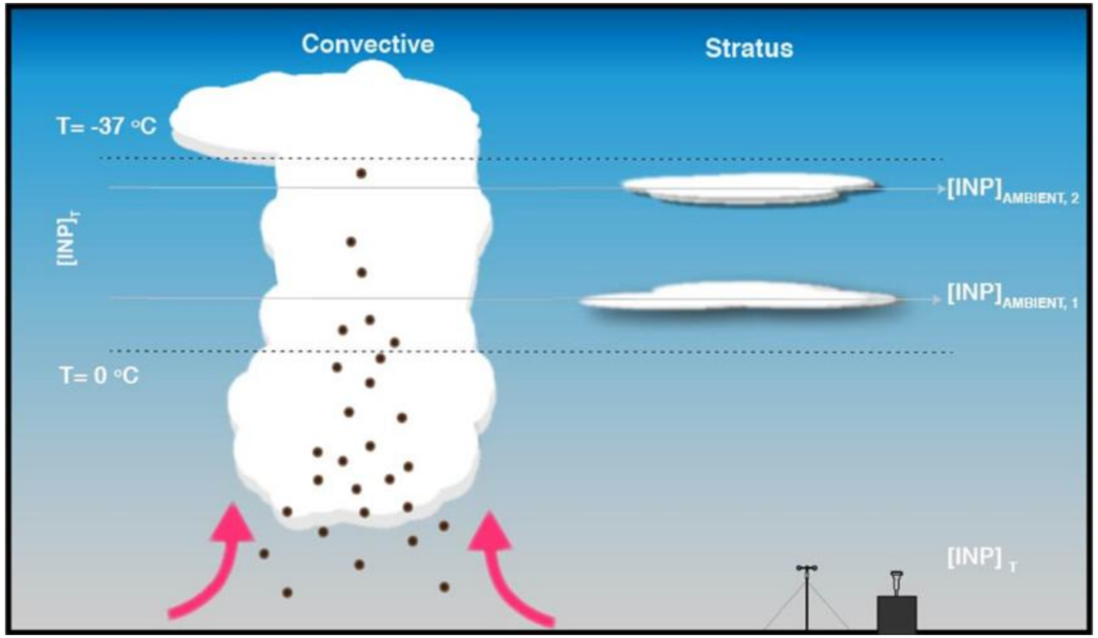
They generally don't have research questions of their own - they develop the computer tools to help other people to do cool things. They might add features to existing software, clear out bugs or build something from scratch. But they don't just sit in front of a computer and write code.

They have to be good communicators who can embed themselves in a team.

Nature: May 2022

Examples of RSE projects



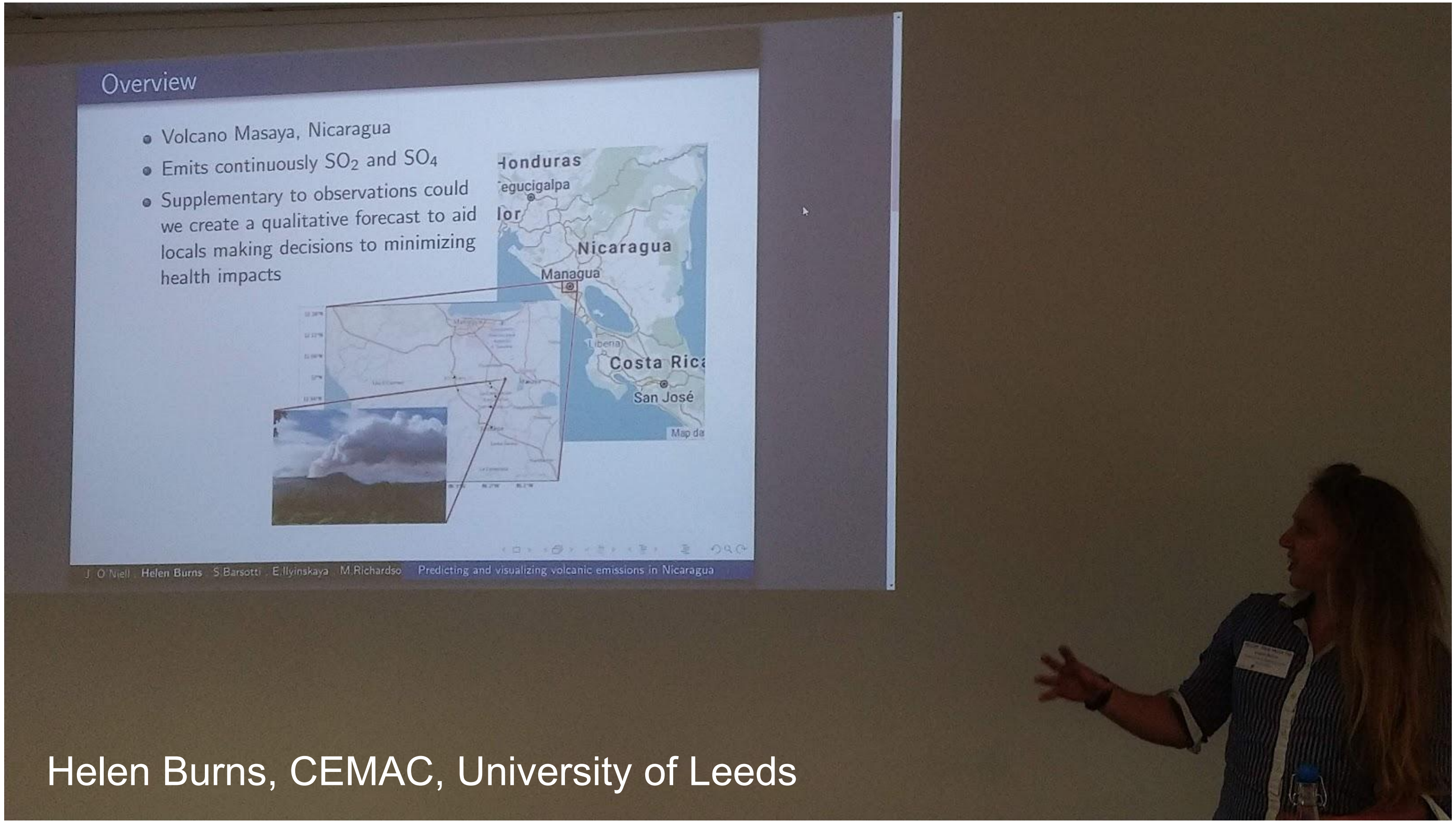


Contribution of feldspar and marine organic aerosols to global ice nucleating particle concentrations

Mar 2017, ACP, Jesús Vergara-Temprado

Examples of RSE projects

Examples of RSE projects



Helen Burns, CEMAC, University of Leeds

Teaching and Outreach

Ten reasons to be a research software engineer

1. Right at the cutting edge of science
2. Travel the world
3. Use the tools and languages that are best suited to the job
4. Open to open source
5. Wall-to-wall bright people
6. Flexibility
7. Why work in only one discipline?
8. Important work
9. Collaboration
10. Choose your own path

Software Sustainability Institute
https://www.software.ac.uk/blog/ten-reasons-be-research-software-engineer

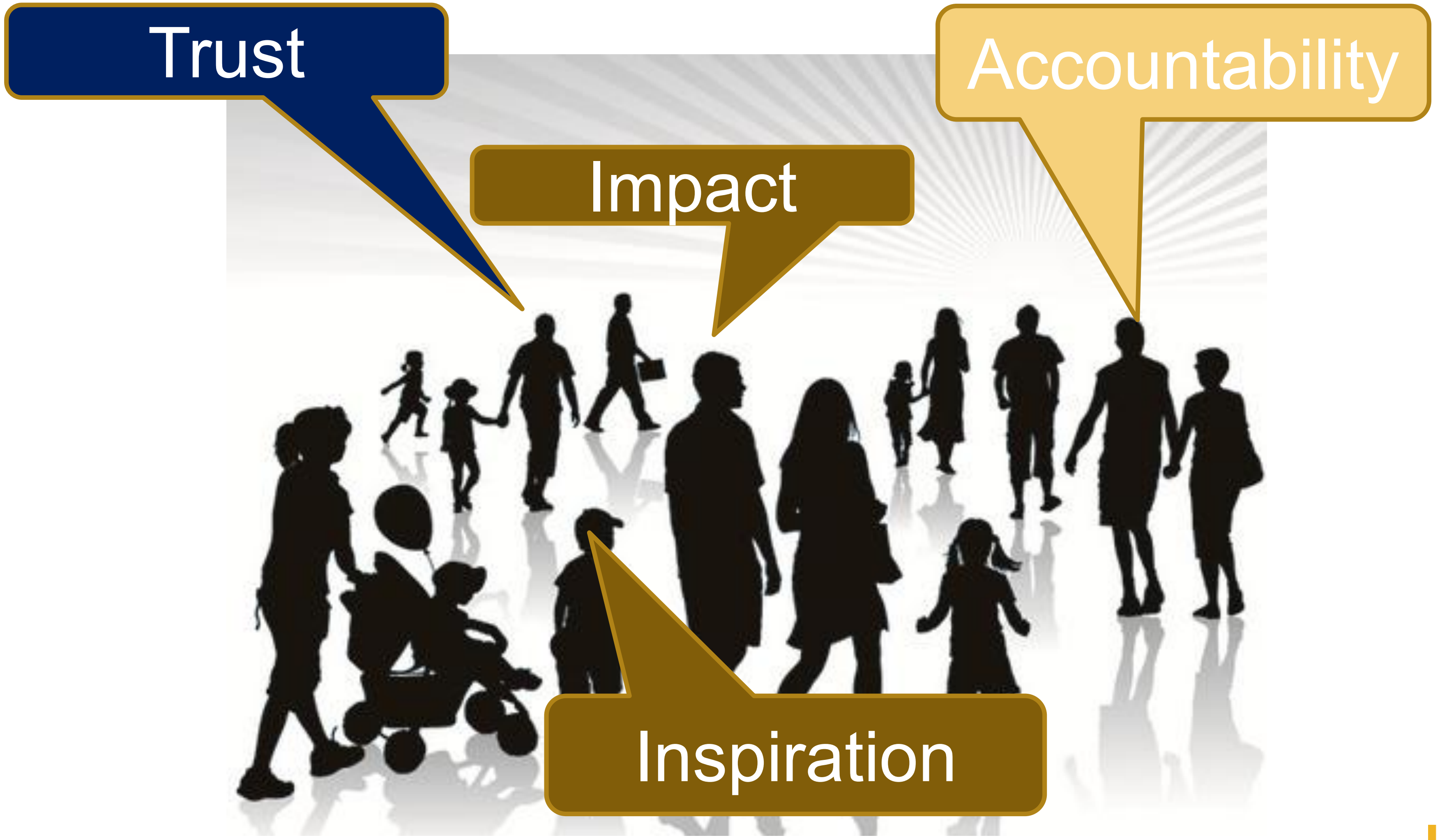
What is public engagement?

- Public engagement describes the myriad of ways in which the activity and benefits of higher education and research can be shared with the public.

- Engagement is a two-way process, involving interaction and listening, with the goal of generating mutual benefit.

NCCPE's definition of Public Engagement

Benefits of public engagement?

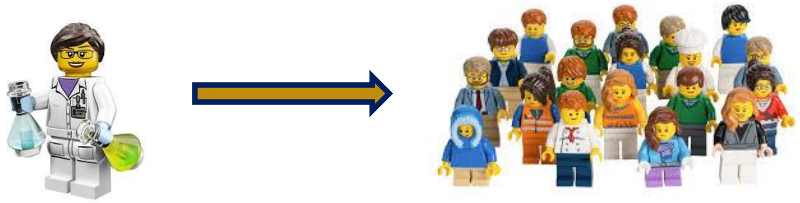


Why are research software skills needed for public engagement?

Development of public engagement in the UK

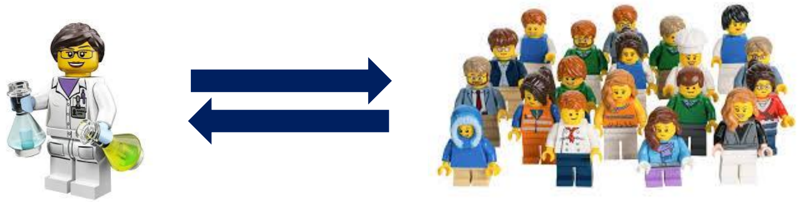
1980s

Deficit Model



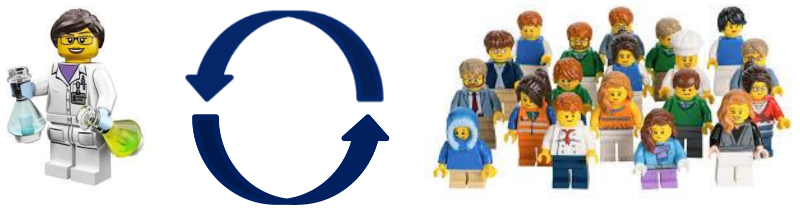
2000s

Dialogue Model



2010

Participatory Model

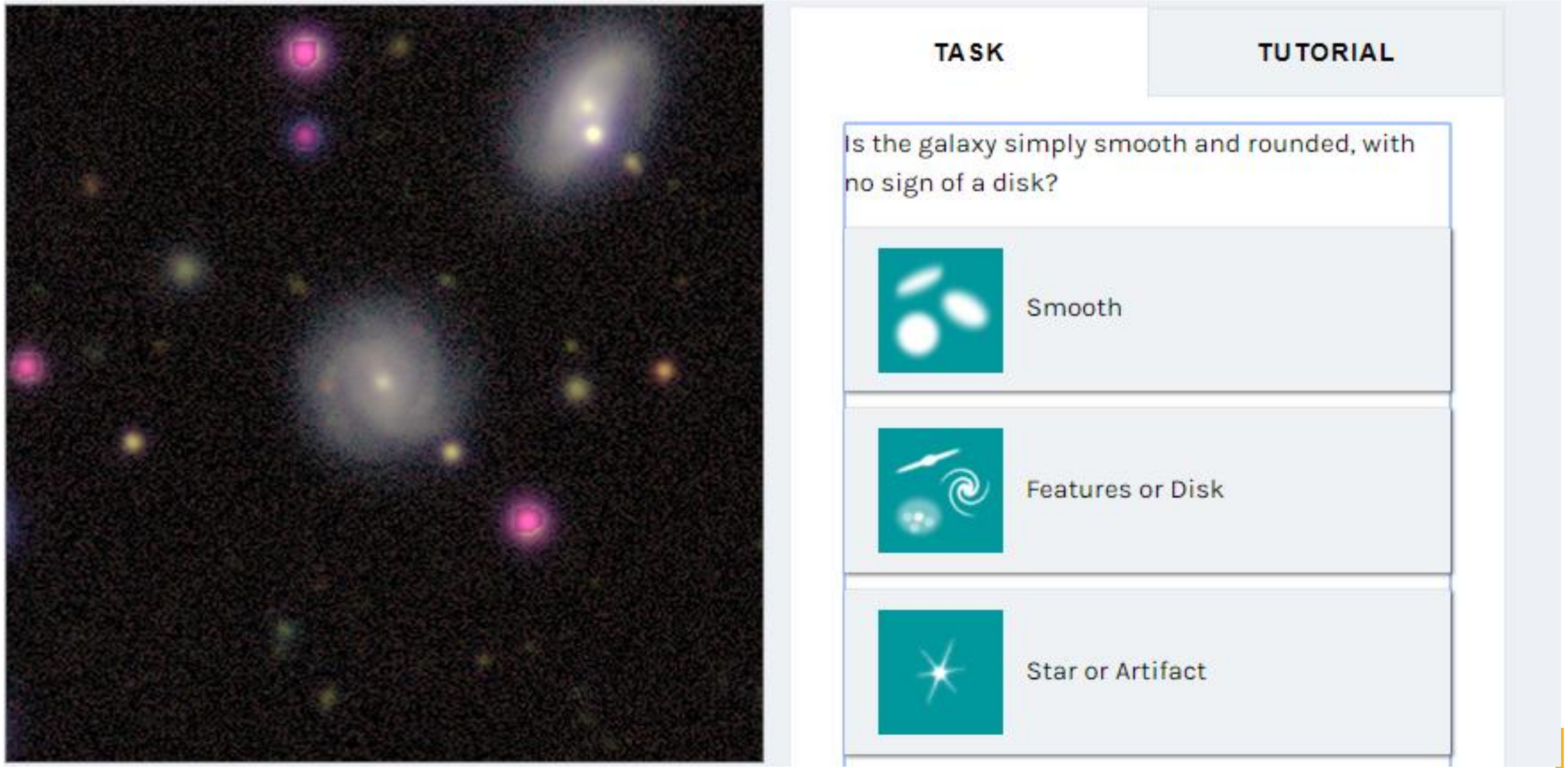


Adapted from: Concannon, Claire & Grenon, Muriel. (2016). Researchers: Share your passion for science!

People powered research

ZOONIVERSE

https://www.zooniverse.org/



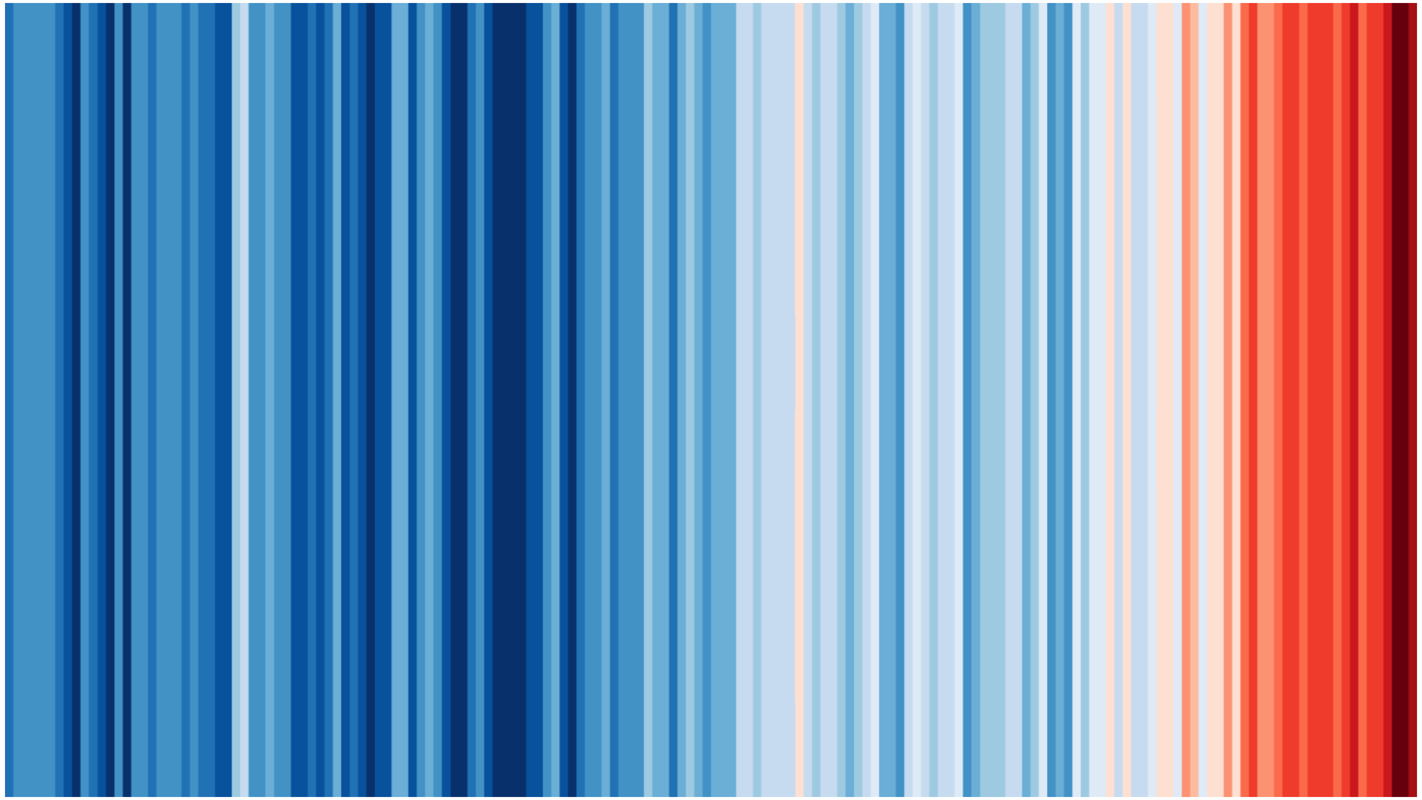
Making data collection easy

Mobile Apps

#BritainBreathing

Web Platforms

Bringing data to life - visualization



Summary

- Research Software Engineering is a new, interesting career option.
- Use your software skills to enable world leading research.
- Public Engagement is increasingly important for research institutions.
- Where possible, public engagement should be a two-way dialogue.
- The technical and coding skills RSEs have offer huge potential for PE.
- Also it’s good fun!